

# Agile AI Agentic Neural Based Framework For Complex Workflow Forming, With Multiple Implementations Upon Upstream And Downstream Scenarios

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## Abstract

The oil&gas sector increasingly depends on agile, reliable, and explainable workflow automation to support production and operations activities such as wellbore fault diagnostic, maintenance, and pipeline scheduling. However, existing genAI-based agents technologies are limited by complex workflow forming, lack of cross-scenario generalization, and insufficient continuous refinery mechanisms, which restrict their ability to support long procedure workflows or adapt to evolving operating conditions. This study proposes an Agile Agentic Neural (AAN) framework that autonomously forms, executes, and refines complex workflows. The approach is designed to address upstream wellbore workovers and downstream pipeline scheduling, enabling event sensing, task planning, tools coordination, and continuous improvement through dual human-machine feedforward mechanism. The objective is to establish an operationally robust and scalable approach that accelerates production decision-making while maintaining interpretability and safety across integrated asset operations.

The AAN framework adopts a neural network inspired architecture with 3 layers. The input layer performs scene recognition, semantic understanding, and task decomposition from multimodal field data (e.g., SCADA, production logs), using distributed self supervised distributed agents to ensure quality-grounded fusion. The computational layer organizes sub-agents within dynamically selected topologies—mesh, tree, code or workflow structures based on operational context. A supervisor agent uses in-context refinement and historical success logs to optimize decisions such as diagnostics, workflow planning, and response actions examining. The output layer assembles executable workflows and employs a dual feedforward loop that integrates engineer feedback and machine-observed metrics, enabling digital-twin-like self-refinement without retraining. Validation includes 60 shale wellbore maintenance cases (pump leaks, rod failures, scaling, wax deposition) and pipeline scheduling scenarios within previous couples of years, involving equipment faults, weather anomalies, and supply-demand disturbances. Ablation studies compare four feedback mechanisms to assess robustness.

Across wellbore maintenance tasks, the AAN framework achieved human and agent evaluation scores of 0.76-0.98, with standard deviations <0.1 in event sensing, task planning, agent coordination, and tool use. Pipeline scheduling cases showed

Access the latest version of skill specification of agile workflow system:<https://github.com/HChen5965/Agileworkflow-APOGCE-26> (Skills only)

comparable performance, with 0.74- 0.93 accuracy and strong stability under multiphase flow and hydrat risk conditions. Cross-scenario generalization maintained 0.85-0.94 accuracy, though toolusage precision declined in highly heterogeneous cases. Ablation testing confirmed the superiority of the global dual-feedback design, achieving 0.78-0.98 accuracy with the lowest variance (0.02-0.19), highlighting its value for real-time production control and risk mitigation.

**Keywords:** Wellbore & Pipeline maintenance&decision excellence ; Production & operational excellence ; Autonomous workforce forming; Generalizable implementations; AI driven method.

## 1. Introduction

The genAI transformation targeted across upstream and downstream oil and gas scenarios predominantly necessitates the re-engineering of a series of complex processes and the integrated application of business and data to accomplish decision-making, planning, and execution within intricate scenarios (Wang et al. 2025). With the continuous accumulation of data resources, the increasing automation of workflows, sustained improvements in asset reliability, and the ongoing enhancement of decision-support tools, contemporary genAI applications have evolved beyond classical single-scenario question-answering. They now extend into exploring and discovering global optimal within production and management processes, enabling the generation and execution of autonomous workflows based on complex decision intents (Chen et al. 2024). Against this backdrop, classical approaches, which rely solely on a foundational genAI model and existing technical pathways, are limited. They can only achieve content generation and semantic understanding for specific, isolated scenarios. These approaches lack the capability for accurate planning of complex task flows and cannot generalize across different scenarios through tool invocation or semantic interface access to execute complex processes.

With impact spans deep thinking mechanisms (Deepseek, 2026), content synthesis, professional knowledge retrieval (Edge & D, 2024), complex logical inference and tool utilization (Schick et al. 2023; Schmidgall et al. 2025; Krishnan, 2025), task execution (Feng et al. 2025), and reflective optimization (Ocker et al. 2025). These advances have collectively facilitated the emergence of near real-time, interpretable decision-support systems, marking a significant evolution in interactive analytics. However, prevailing generative AI implementations predominantly cater to scenarios characterized by deterministic work processes and well-defined procedural boundaries. A significant limitation lies in their inherent lack of mechanisms for agile workflow creation and continuous, self-refining adaptation in response to novel or dynamically evolving constraints, thereby establishing trustworthy AI regrading complex tasks (Schmidgall et al. 2025) This capability gap primarily stems from underdeveloped frameworks for inter-agent collaboration and human-agent cooperation (Krishnan, 2025; Feng, 2024; Ocker, 2025). These inadequate frameworks consequently hinder multi-agent systems, or 'swarm agents,' from achieving the autonomous, iterative improvement necessary for tackling complex, long-processing and fuzzy scenarios such as decision-making challenges. Specifically, the absence of flexible thought-chaining and reasoning architectures, coupled with a lack of closed-loop feedback mechanisms from final output terminals, restricts the ability of these systems to dynamically adjust their operational logic and task execution pathways. This limitation ultimately curtails their efficacy and reliability in dynamic, long-process multi-hops AI implementations where continuous adaptation is paramount for success, which significantly influence the long term usage of AI-native products.

To address these challenges, genAI based agentic neural networks incorporating feedback loops and adaptive weight factor adjustments have been developed to enable self-refinement in frequently encountered tasks such as mathematical computation, creative writing, and adaptive machine learning (Ma et al. 2025). Furthermore, inspired by Monte-Carlo hierarchical structures, these systems utilize prompts to represent nodes, agentic neural networks as operators, and mixture-of-expert (MoE ) to model graphs and AFlow neural architectures, respectively (Zhang et al. 2025). The operation of complex integrated energy IT systems, which fall under the category of human-machine collaborative auxiliary decision-making problems, is often subject to multi-factor disturbances. Conventional neural-layer architectures struggle to achieve comprehensive coverage and maintain robust generalization across diverse oil and gas applications, especially at the beginning of workflow managing. Drawing inspiration from LLM-aided ontology engineering, which facilitates precise workover designs and scenario replication (Arabi et al. 2025; Mai et al. 2024), as well as building upon prior advancements in agentic neurals, it is reasonable to posit that a swarm agentic based framework could offer significant contributions. Such an approach, characterized by architectural flexibility and self-refinement capability, holds considerable potential for the effective "colonization" or application across a wider range of integrated energy industry processes.

The oil and gas sector possesses a substantial accumulation of knowledge and experience. Utilizing an MoE architecture allows for leveraging these historical insights in complex analysis, diagnosis, and decision-making (Du et al. 2025), while also achieving a better balance between computational overhead and model refinery. Consequently, for complex task chains that require linking multiple actions to execute a complete process, a common approach involves employing an MoE system composed of foundation models, agents, specialized software, and expert knowledge bases. This aims to holistically balance computational cost, reliability, knowledge reuse, and intelligent user experience. Such multi-model, multi-system collaborative MoE systems have been successfully applied in various upstream and downstream scenarios, including exploration and pipeline transportation (Chen et al. 2024 & 2025) .

Nevertheless, the nature of genAI, which relies on semantic understanding, content retrieval, and probabilistic prediction to enhance information acquisition and utilization efficiency (Vyhmeister et al. 2026), introduces specific challenges for industrial applications. The characteristics of natural language human-computer interaction and content generation mean that generative agents can produce content biased towards user preferences based on contextual memory. In industrial settings, this may lead to output drift or convergence to local optima for specific users after deployment (Vyhmeister et al. 2026). For long-chain processes and complex decision-making tasks in oil and gas, such drift and local optima present risks to the robustness of application systems (Confalonieri,&Guizzardi 2023; Mai et al. 2024). To address these operational challenges, it is worth to develop agent self-refinery techniques tailored to the characteristics of the energy industry. This involves using application feedback loops for reinforcement learning to enable autonomous improvement for multi-objective, long-process complex tasks, aiming for post deployment self-refinement. Drawing inspiration from the feedforward mechanism in neural networks ( Hu et al. 2024; Ma et al. 2025), this manuscript proposes an agentic neural architecture based on semantic memory and dual human-machine feedback. For specific scenarios addressable through tool invocation and workflow assembly (Schick, 2023), this architecture utilizes linguistic feedback streams to autonomously adjust the network nodes and configuration of the agentic neural system, thereby enhancing its robustness in the face of subtle changes in specific scenarios and requirements (Chen & Jin et al, 2025). However, for most of cases already known, they showed to pay more attention on adaptive refinery regarding to the applications with fixed workflow, but put less consideration on elastic workflow which might driven continuous reformation of oil and gas works.

To overcome the challenges regarding to agile workflow generation and self-refinery in complex oil and gas scenarios, this manuscript proposes an agentic neural based framework with autonomous evolution and agile workflow generation capabilities, referred to as the AAN framework. The AAN introduces an innovative human-machine dual-feedforward architecture and a generation algorithm for elastic network structures within neural network hidden layers (including layers, network, hierarchy, coding, etc). This enables the framework to proactively and flexibly select network architectures according to specific implementations, thereby facilitating agile workflow generation. The proposed framework was validated using historical data from 60 shale oil wellbore downhole operation scenarios and 3 transnational natural gas transmission pipelines spanning Central-Asia to East-Asia. The validation focused on wellbore downhole operations and pipeline gas transmission scheduling decision-making tasks, assessing the AAN's capabilities in terms of agile workflow generation, scenario reuse, and autonomous evolution. The manuscript is structured as follows: Section 2 illustrates an instance of self-refinery agentic mechanism, thereby delivering direct insight regarding to the benefit of in-context based continuous improvement of complex workflow tasks; Section 3 details the AAN framework, emphasizing its agile workflow construction mechanism and elastic network selection ability. Section 4-5 presents experimental verification and discussion based on past years of historical data from the 60 shale oil wellbores and 3 years' transnational pipelines operational logs. Conclusion and discussion are given in Section 6.

## **2. Instance of self-refinery mechanism**

Fig 1 illustrates an example of an agile workflow with autonomous evolution capabilities, constructed based on the proposed AAN described in this manuscript. The left and right parts of the figure depict the initial occurrence of a scenario and the agent's self-refinery process when an identical or similar scenario reoccurs, respectively. During the initial occurrence, the AAN executes the full agent task chain to achieve autonomous workflow completion. This approach aligns with case based reasoning methods where workflows are generated from scratch based on predefined structural adaptations. When a similar or identical scenario reappears, the AAN first retrieves and reuses historically successful execution cases, significantly reducing reasoning time through experience reuse. Concurrently, building upon these historical successes, the AAN utilizes its self refinery capability to identify potential improvements for the current scenario and constructs a refined, agile workflow, thereby achieving the system's capability for autonomous optimization through semantic pre-feed information flow. The architectural details enabling this agile process are elaborated within the next coming sections.

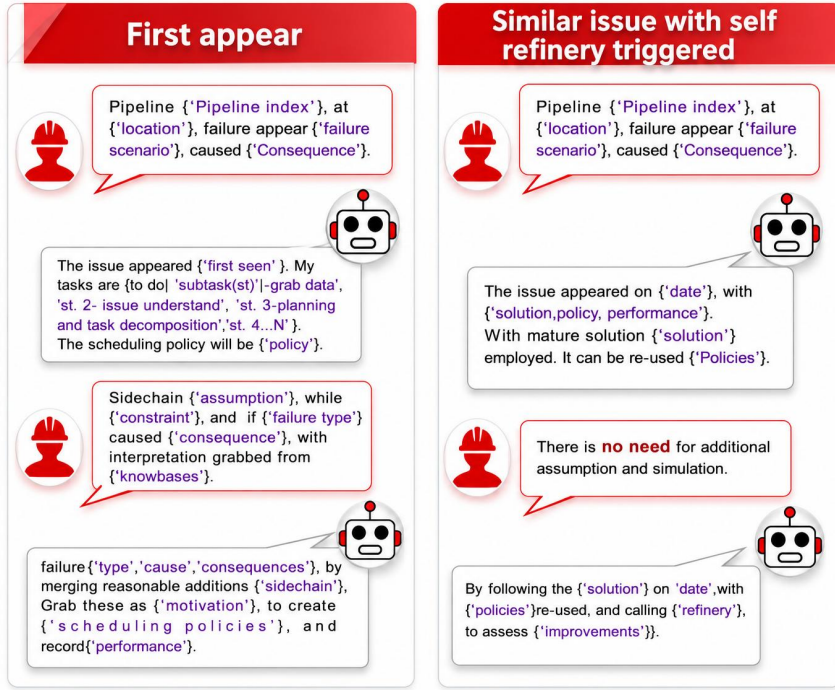


Fig 1—Agile workflow with self-refinery ability: an instance

### 3. Mathematical representation of AAN

The AAN is structured as a classical 3 layers agentic neural architecture, comprising an input Layer, a computational layer, and an output layer, designed for dynamic workflow formation and optimization. This architecture enables the system to process multimodal inputs, decompose complex tasks, and generate validated workflows through a continuous feedback loop. The overall procedure is represented by algorithm 1.

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#### Algorithm 1: Construction of AAN for Agile Workflow Generation and Optimization

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##### // Input Layer Processing

**Input:** Multimodal decision supporting data stream  $X=[X_1, X_2, \dots, X_m]$  (including text, speech, image, signals, etc.)

**Output:** Validated workflow  $Y$  with rationality score  $r \geq \gamma$

##### 1: // Input Layer Processing

2: Intent Understanding:  $z_{intent} = \Phi_{intent}(X) \rightarrow$  Validate via self-consistency (e.g., voting).

3: Task Decomposition:  $\{\tau_1, \dots, \tau_K\} = \Phi_{decomp}(z_{intent}) \rightarrow$  Build task graph  $G=(V, E)$ .

4: Multimodal Fusion:  $X_{fused} = \Phi_{fuse}(X) \rightarrow$  Monitor data quality via task observer.

##### 5: // Computational Layer Processing

6: Initialize distributed network: Supervisor agent S, sub-agent cluster  $\{A_1, \dots, A_K\}$ .

7: for each sub-task  $\tau_k$  do

8: Assign model  $\theta_k = \Phi_{model}(\tau_k)$  to sub-agent  $A_k$ .

9: Execute:  $y_k = \Phi_{exec}(X_{fused}, \theta_k) \rightarrow$  Optimize  $\theta_k$  via in-context-refinery (historical logs  $H$ ).

10: end for

11: Perform dynamic orchestration: Adjust graph topology  $G$  (e.g., mesh  $\rightarrow$  tree).

##### 12: // Output Layer Processing

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13: Generate Workflow:  $Y = \Phi_{output}(\{y_1, \dots, y_K\})$ .
14: Verify: Compute rationality score  $r = \Phi_{verify}(Y)$ .
15: while  $r < \gamma$  do
16:   Collect human feedback  $f_h$  and machine feedback  $f_m$ .
17:   Compute combined feedback  $f = \alpha f_h + (1 - \alpha) f_m$ .
18:   Adjust network:  $\Delta G = \Phi_{feedback}(Y, f) \rightarrow$  Dynamically modify G via AIGC coding.
19:   Regenerate workflow Y and score r(line 13-14).
20: end while
21: return Y

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**Tab 1—The overall procedure of proposed AAN approach**

The technical architecture of AAN is given in Fig 2, with the main functions of three layers obey the algorithm 1 are determined as follow:

(1) Input layer for multimodal understanding and task decomposition

The input layer ingests a heterogeneous, multimodal data stream  $X$ . Its primary functions are scene recognition, semantic understanding, and task decomposition. Intent understanding is modeled as  $z_{intent} = \Phi_{intent}(X)$ , where the result is validated for robustness using a self-consistency mechanism (e.g., majority voting across multiple samples). The inferred intent is then decomposed into a set of sub-tasks  $\{\tau_1, \dots, \tau_K\}$ , which are structured as a directed acyclic task graph  $G = (V, E)$ . Input agents, organized under a self-supervised framework (Arabi, et al. 2025), act as task observers. They perform multimodal fusion  $X_{fused} = \Phi_{fuse}(X)$  and continuously monitor data integration quality, enabling a self-refining data preparation process.

(2) Computational layer for distributed neural execution

The computational layer orchestrates sub-task execution via a distributed network of agents, inspired by the a set of former agentic neural-like architecture. A supervisor agent manages a cluster of sub-agents  $\{A_1, \dots, A_K\}$ , each dedicated to a specific sub-task  $\tau_k$ . For each sub-task, an appropriate base model (e.g., a large-scale foundation model) and a network topology (e.g., mesh, layer) are selected from a model pool, formalized as  $\theta_k = \Phi_{model}(\tau_k)$ . Sub-agent execution is defined by  $y_k = \Phi_{exec}(X_{fused}, \theta_k)$ . Crucially, each agent employs an in-context-refinery process, utilizing a historian log of successful cases to refine its reasoning and output without updating the foundation model's weights. The supervisor agent dynamically adjusts the connectivity of the task graph to optimize parallel execution, effectively forming a swarm agent cluster tailored to the problem.

(3) Output layer for workflow generation and dual-feedback refinement

The output layer synthesizes the results from the computational layer into a coherent workflow  $Y = \Phi_{output}(\{y_1, \dots, y_K\})$ . A key component is the verification function  $r = \Phi_{verify}(Y)$ , which produces a rationality score. If  $r$  falls below a threshold  $\gamma$ , a dual-feedback mechanism is engaged. Feedback is collected from both human experts ( $f_h$ ) and automated metrics ( $f_m$ ), and combined as  $f = \alpha f_h + (1 - \alpha) f_m$ . This feedback is processed by the supervisor agent to compute adjustments  $\Delta G = \Phi_{feedback}(Y, f)$  to the task graph. These adjustments, often expressed in natural language, are translated into structural modifications to the AAN via AIGC coding, creating a closed-loop, self-refinery system.

(4) AAN system training and self-refinery process

The AAN framework is trained without fine-tuning the underlying foundation models. Instead, it leverages a continuous refinery scheme (Chen & Jin et al. 2025) applied to a historian log of successful workflows to gain and test prompt variants for complex, multi-hop tasks. The system continuously refines through the previous mentioned dual-feedback loop. The supervisor

agent uses external domain knowledge and the historian log to enhance intents decomposition, tools using, and predict outcomes. This multi-layer collaborative architecture enables the AAN to achieve adaptive assembly, generalize across multi-hop tasks, and establish a continuous optimization mechanism based on execution deviation, resulting in end-to-end autonomous decision-making from task semantics to execution effects.

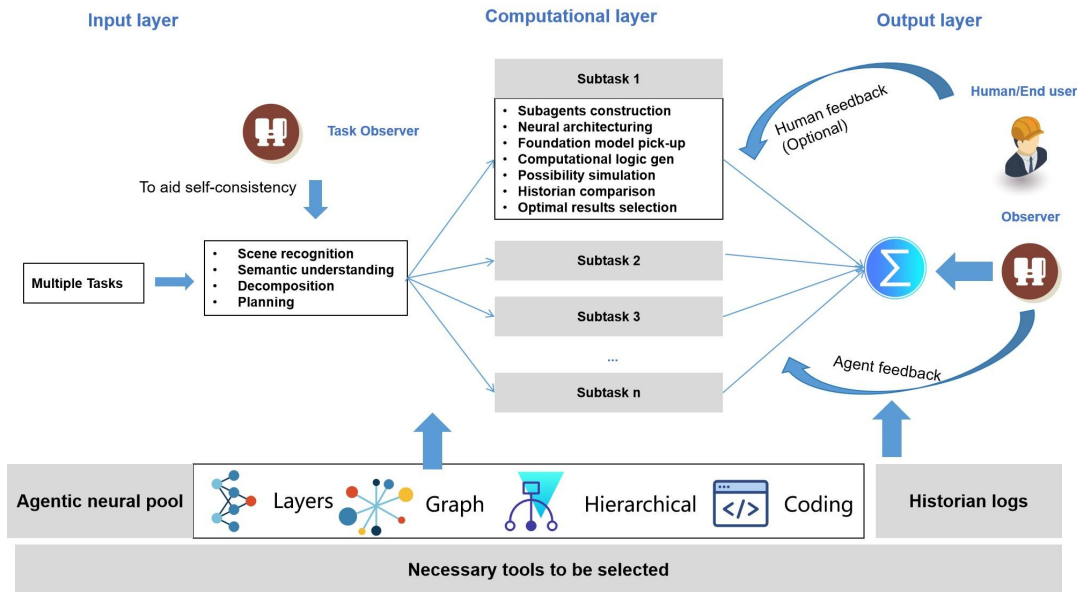


Fig. 2— Technical architecture of AAN proposed

#### 4.Experimental Design

To validate the agile workflow generation capability of the proposed AAN architecture and its cross scenarios generalizability, this study selects two cross-value-chain scenarios (one upstream and one downstream, respectively), namely, wellbore downhole operations for shale oil, and long trans-national pipeline scheduling for evaluation. The experiments assess the AAN's performance in event sniffing, task planning, agent liaison, tool using, agile workflow generation, as well as continuous refinery, in order to manage the whole workflow ability. The specific experimental designs for these two scenarios are as follows:

The downhole operations scenario focuses on the operational excellence purposes of 60 shale oil wellbores located in major oilfields in Western China. For wellbore failures such as pump leakage, rod breakage, scaling, and wax deposition, the study verifies the ability of the AAN to generate adaptive workflows for different wells. The workflow generation tasks include fault event triggering, fault diagnosis, workover strategy combination, effectiveness and economic benefit evaluation, and feedback for optimization suggestions. The long distance trans-national pipeline scheduling scenario involves decision-making events for 3 transnational natural gas transmission pipelines traversing Central and Southeast Asia, with a total distance over 6500 km. The experiment evaluates the AAN's capability to perform agile workflow orchestration in response to triggers unexpected events, such as sudden equipment failures, human factors, climate and meteorological conditions, fluctuations in gas supply/demand, commodity price volatility, and end-user sales demand changes.

In terms of baseline data and external knowledge bases, for wellbore downhole operations, baseline data includes production, operational, and historical maintenance records from 60 CNPC shale-oil wellbores between 2021 and 2024, complemented by external knowledge from public manuscripts, global patents, and historical wellbore management system data. This knowledge base aids AAN in intent understanding, process interpretation, and result comparison, enabling swarm agents to form accurate

workflows. Maintenance types covered include pump leakage, rod breakage, scaling, wax deposition, and routine maintenance. Without fine-tuning or retraining, swarm agents retrieve and semantically compare historical cases—including prompts, chains of thought, plans, and outcomes—to learn and adapt execution strategies suitable for the scenario.

For transnational pipeline scheduling, operational data (2020–2024) from 3 selected pipelines, including PPS, SCADA, MES, and ERP systems—serves as the baseline. External knowledge incorporates historical extreme weather events, abnormal climate incidents, scheduling anomalies, accidents, and geopolitical and political-economic environments. A similar swarm agent learning approach, as used in wellbore operations, facilitates autonomous generalization of the agentic neural network.

In terms of conventional experiment setup, after in-context based training, a scenario simulation agent is designed for both wellbore operations and pipeline scheduling tasks. Using historical data and adaptive prompt engineering techniques (Mai et al 2024), 30 events and corresponding reference workflows are generated for each scenario. Each case includes dimensional references such as historical execution records, recent performance evaluations, reference solutions, and standardized benchmarks for task planning and execution. To test the AAN's generative and self-refinery capabilities, each of the 30 cases comprises 3 temporally sequential occurrences of the same event (e.g., a pipeline leak occurring at different times with varying maintenance histories), along with their solutions and explanations. This design evaluates the AAN's capacity for temporal self-organization and self-optimization.

The generated outcomes are assessed based on event sniffing and semantic understanding, task planning, agent orchestration, tool selection strategy, workflow agility (elasticity), and self-refinement. Since the study focuses on evaluating the agile workflow assembly capability of the agentic neural, the primary assessment examines whether the network successfully generates agile workflows. The outputs are in natural language, and both automated agent evaluation for objective and human evaluation for subjective may contain errors. Given that human evaluation carries greater weight, the results combine scores from the simulation agent and human evaluators. Both agent and human scores range from 0 to 1 (retaining two decimal places) to determine accuracy. For the 30 experiments, the average accuracy and standard deviation are calculated for both agent and human evaluations, reflecting overall accuracy and robustness.

In specific, the experiments for wellbore downhole operations and pipeline scheduling agile workflow assembly follow these steps: For the main performance testing, (1) The user inputs an intent command via a dialogue interface, triggering the AAN to initiate workflow generation. This involves stages of reasoning, task planning, workflow generation, tool using, agile workflow generation, and continuous optimization. Based on the generated workflow, an appropriate AAN network architecture is selected as the execution framework (Network, layer, graph, or coding etc. based on actual application scene). (2) Role-specific agents retrieve references from historical execution events and performance evaluations to identify suitable benchmarks for self-refinery of the steps in (1). (3) Computational-layer agent clusters simulate 3 potential self-refinery schemes based on historical events and evaluation conclusions. These are compared with the self-improvement strategies from (2), and the optimal scheme is autonomously selected. (4) The self-evolve outcome undergoes automatic verification via observers within the AAN. If the feedback is positive, the process proceeds; if uncertain or negative, step (3) is repeated. (5) The final result is output and evaluated by both an evaluation agent and the end-user. The evaluation involves comparison with reference solutions.

To verify the scenario generalizability of the AAN, 20 events are randomly selected from the combined datasets of the two scenarios (wellbore downhole operation and pipeline scheduling decision making). These include 3 temporally sequential

occurrences of the events. The aforementioned 5 experimental steps are repeated, and the global average accuracy is computed. The evaluation employs the same human-agent collaborative scoring mechanism.

To assess the robustness of the AAN architecture in agile workflow generation, an ablation test is conducted: 20 event sets, randomly mixed from wellbore maintenance and pipeline scheduling scenarios, are selected. The impact of different feedback designs on the cross-scenario performance of the agentic neural architecture is tested through comparative experiments. The comparisons include configurations such as global feedback (dual), global feedback (without human), dual feedback to input only, and feedback to input without human response. Each configuration is evaluated on scene perception, task planning, and feedback learning performance, using the same human-agent collaborative scoring mechanism. The results and discussion will be delivered in the next section.

## 5 Experimental results

The experimental results regarding to wellbore maintenance, and pipeline scheduling decision making, scene generalization, as well as ablation testing, are illustrated in tables 2-6, respectively.

| Application scene/Category | Human evaluation (Average) | Human evaluation (standard deviation) | Agents evaluation (Average) | Agents evaluation (standard deviation) |
|----------------------------|----------------------------|---------------------------------------|-----------------------------|----------------------------------------|
| Events sniffing            | 1                          | 0.088                                 | 1                           | 0.09                                   |
| Task planning              | 0.76                       | 0.088                                 | 0.93                        | 0.09                                   |
| Agents liaison             | 0.85                       | 0.088                                 | 0.945                       | 0.09                                   |
| Tool using                 | 0.92                       | 0.088                                 | 0.99                        | 0.09                                   |
| Agile workflow             | 0.89                       | 0.088                                 | 0.78                        | 0.09                                   |
| Continuous refinery        | 0.98                       | 0.088                                 | 0.81                        | 0.09                                   |

**Table 2—Performance statistics of proposed methodology for wellbore downhole maintenance**

Based on the results present in table 2, the proposed AAN approach demonstrates satisfactory performance in the upstream wellbore maintenance. Both agent and human evaluations generally yield scores above 0.75, with standard deviations below 0.1. This indicates that the agentic neural system exhibits considerable robustness in elastically generating workflows for different wellbores and at varying time points, particularly in key aspects such as event sniffing, task planning, agent liaison, tool usage, agile workflow formation, and continuous self-refinery. Regarding task planning, a notable discrepancy is observed between human evaluation results and the agent's self-evaluation scores. This discrepancy primarily stems from limitations in the prompt design, which sometimes restrict the agents' ability to fully adapt to time-dependent changes in wellbore status resulting from ongoing operations and maintenance activities. Consequently, reliability is primarily ensured for the main workflow branches. However, this limitation is mitigated by the AAN's dual feedforward mechanism incorporating a human-in-loop (HIL) design. This architecture allows for compensatory control of task planning errors through semantic pre-feed information flow, enhancing the system's overall adaptability and accuracy .

| Application<br>scene/Category | Human evaluation<br>(Average) | Human evaluation<br>(standard deviation) | Agents evaluation<br>(Average) | Agents evaluation<br>(standard deviation) |
|-------------------------------|-------------------------------|------------------------------------------|--------------------------------|-------------------------------------------|
| Events sniffing               | 1                             | 0.1                                      | 0.93                           | 0.053                                     |
| Task planning                 | 0.76                          | 0.1                                      | 0.91                           | 0.053                                     |
| Agents liaison                | 0.91                          | 0.1                                      | 0.88                           | 0.053                                     |
| Tool using                    | 0.85                          | 0.1                                      | 0.82                           | 0.053                                     |
| Agile workflow                | 0.74                          | 0.1                                      | 0.86                           | 0.053                                     |
| Continuous<br>refinery        | 0.93                          | 0.1                                      | 0.79                           | 0.053                                     |

**Table 3—Performance statistics of proposed methodology for pipeline scheduling decision policy making**

In the domain of pipeline scheduling decision making, the proposed methodology demonstrates agile workflow assembly capabilities comparable to its performance in the upstream use case, and exhibits robust competence in sustained self-refinery facilitated by its dual feedforward mechanism. Although the subjective evaluation scores for task planning and agile workflow generation are relatively lower compared to those in the wellbore maintenance implementations, they still maintain an accuracy exceeding 70%. This can be attributed to the inherent complexity of pipeline scheduling, which constitutes a multi-objective collaborative optimization and strategy generation problem influenced synergistically by multiple external factors (Chen& Jin et al. 2025), thereby introducing greater variability and stricter criteria in human assessment. Owing to the effective implementation of continuous self-refinement (enabled by HIL feedforward behaviors) within the AAN architecture, the performance on these aforementioned metrics can be enhanced through ongoing optimization, ultimately leading to the generation of personalized solutions tailored to different end-users.

In terms of scene generalization, the AAN demonstrates robust generalization capabilities across the 20 above mentioned randomized mixed-scenario tests, with overall statistical results indicating satisfactory performance. However, due to the randomness of scenario selection, a moderate decline in accuracy was observed concerning tool using within lengthy workflows and the effectiveness of continuous self-refinement. This can be largely attributed to the significant differences in process complexity and the inherent disparity in business logic between the upstream (wellbore equipment maintenance) and downstream (cross-value-chain pipeline scheduling decision-making) scenarios, which impose more demanding requirements on the generalization capacity of the AI system. This observation further underscores the necessity of the human-AI collaboration mechanism inherent in the proposed AAN architecture.

| Scene<br>Generalization | Human evaluation (Average) | Human evaluation<br>(standard deviation) | Agents evaluation (Average) | Agents evaluation<br>(standard deviation) |
|-------------------------|----------------------------|------------------------------------------|-----------------------------|-------------------------------------------|
| Events sniffing         | 0.85                       | 0.089                                    | 0.9                         | 0.066                                     |
| Task planning           | 0.91                       | 0.089                                    | 0.91                        | 0.066                                     |
| Agents liaison          | 0.94                       | 0.089                                    | 0.9                         | 0.066                                     |
| Tool using              | 0.72                       | 0.089                                    | 0.78                        | 0.066                                     |
| Agile workflow          | 0.88                       | 0.089                                    | 0.85                        | 0.066                                     |

**Table 4—Performance of proposed AAN approach in terms of scene generalization**

In terms of robustness assessment, an ablation study was conducted by comparing the following four feedforward designs to determine the necessity of adopting a dual-feedforward mechanism by testing the following categories:

- For global feedback (dual): Employs dual feedback from both human and agentic information flows. The feedback is simultaneously delivered to both the computational layer and the input layer.
- Global feedback (without human): Utilizes only the agentic information flow for feedback. The feedback is delivered to both the computational layer and the input layer.
- Dual feedback to input only: Employs dual feedback from both human and agentic information flows. However, the feedback is delivered solely to the input layer.
- Feedback to input by agentic: Utilizes only the agentic information flow for feedback, and the feedback is delivered solely to the input layer.

Analysis of the average accuracy and corresponding standard deviations, as illustrated in the figures below, reveals that the global dual feedback mode (human + agent) achieved the best performance in agile workflow generation. The other 3 feedforward strategies resulted in slightly inferior performance either in terms of workflow generation accuracy or greater volatility in the outcomes compared to the global dual feedback mode. This finding underscores the critical role of the HIL mechanism for the effective generation and organization of elastic workflows within this AAN architecture. Furthermore, applying dual feedforward mechanism to both the computational layer and the input layer facilitates cross layer optimization between the 2 computational hierarchies and chains of thought. This dual level feedback mechanism significantly enforces the AAN's performance in handling long-sequence, multi-hop tasks, particularly evident in improvements regarding multi-intent comprehension, lengthy workflow generation, appropriate tool usage, and continuous workflow refinement, where both accuracy and robustness are markedly improved.

| Ablation testing/Category    | Global feedback (dual) | Global feedback(without human) | Dual feedback to input only | Feedback to input by agentic |
|------------------------------|------------------------|--------------------------------|-----------------------------|------------------------------|
| Events sniffing _Average     | 1                      | 1                              | 0.85                        | 0.85                         |
| Task planning _Average       | 0.98                   | 0.98                           | 0.82                        | 0.81                         |
| Agents liaison _Average      | 0.94                   | 0.93                           | 0.77                        | 0.83                         |
| Tool using _Average          | 0.87                   | 0.82                           | 0.76                        | 0.77                         |
| Agile workflow _Average      | 0.78                   | 0.87                           | 0.69                        | 0.72                         |
| Continuous refinery _Average | 0.83                   | 0.76                           | 0.69                        | 0.58                         |

**Table 5—Ablation testing resultss (Accuracy scope)**

| Ablation testing/Category                  | Global feedback<br>(dual) | Global feedback(without<br>human) | Dual feedback to<br>input only | Feedback to input by<br>agentic |
|--------------------------------------------|---------------------------|-----------------------------------|--------------------------------|---------------------------------|
| Events sniffing _Standard<br>deviation     | 0.02                      | 0.03                              | 0.09                           | 0.16                            |
| Task planning _Standard<br>deviation       | 0.08                      | 0.09                              | 0.11                           | 0.21                            |
| Agents liaison _Standard<br>deviation      | 0.12                      | 0.13                              | 0.14                           | 0.24                            |
| Tool using _Standard deviation             | 0.1                       | 0.17                              | 0.18                           | 0.19                            |
| Agile workflow _Standard<br>deviation      | 0.19                      | 0.14                              | 0.18                           | 0.27                            |
| Continuous refinery _Standard<br>deviation | 0.15                      | 0.15                              | 0.22                           | 0.23                            |

**Table 6—Ablation testing resultss (Standard deviation scope)**

## 6 Conclusion and discussion

This study proposed and validated an AAN architecture designed for agile workflow generation in complex oil and gas upstream and downstream scenarios. The core contribution of this research is an agentic framework that effectively addresses the limitations of large foundation models and static multi-agent systems by implementing a dynamic, self-evolving structure inspired by neural network liked principles and enhanced by a HIL dual feedforward mechanism.

The experimental findings lead to the following primary conclusions:

- **Proven efficacy:** The AAN architecture successfully generates and executes agile workflows for both wellbore maintenance and pipeline scheduling tasks, as evidenced by accurate scores and low standard deviations in evaluations. It demonstrates a significant capability in key areas of autonomous workflow forming and executing regarding to the upstream and downstream applications, such as event sniffing, task planning, agent liaison, and continuous self-refinement.
- **Essential HIL:** The incorporation of human expertise via a dual feedforward mechanism is not merely beneficial but critical. The ablation tests confirmed that the global dual feedback mode yields the most accurate and robust outcomes, highlighting the importance of human feedback for compensating for the subtleties and temporal dynamics that pure semantic agents might miss.
- **Effective cross scenario generalization:** The AAN exhibits a commendable ability to generalize across different scenarios within the oil and gas value chain. While performance can vary with the complexity and dissimilarity of the combined workflows, the architecture fundamentally supports the transfer of insights and adaptations between upstream and downstream operations.
- **A path forward for industrial AI:** This work provides a scalable and data driven alternative to full model fine-tuning for long sequence industrial tasks. By leveraging historical data and a structured multi-agent network, the AAN offers a more computationally efficient and interpretable approach to managing complex processes in the oil and gas industry.

The current research drives some future prospects, primarily related to the performance dip observed in highly complex, hybrid scenarios involving long and multi-hop workflows. The reliance on the quality of historical data and prompt design also presents a dependency. Future work will focus on enhancing the AAN's few-shot learning capabilities to reduce this dependency and on refining the feedback compression mechanism to improve efficiency in even more extended and intricate operational sequences. Exploring the application of the AAN to other energy industry segments, such as reservoir simulation or renewable energy grid integration, represents a promising direction for further validating and extending its generalizability.

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## **Conflict of interest**

None.